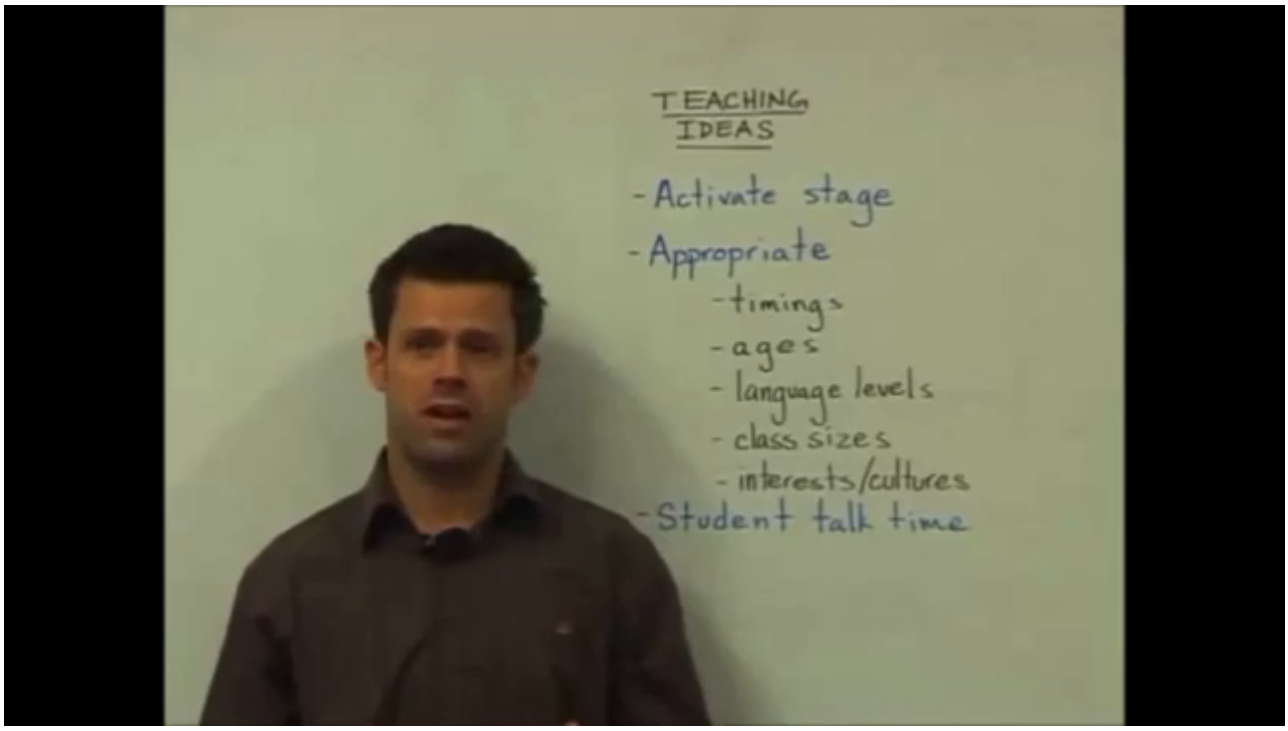
Frame Review: Keep the Lesson

Locally qualified real-lesson release / 23 October 2026
Joseph Ayanda / AI-assisted development
AWS upgrade pending existing-account credential refresh. The public app still serves the earlier authored-lesson release.



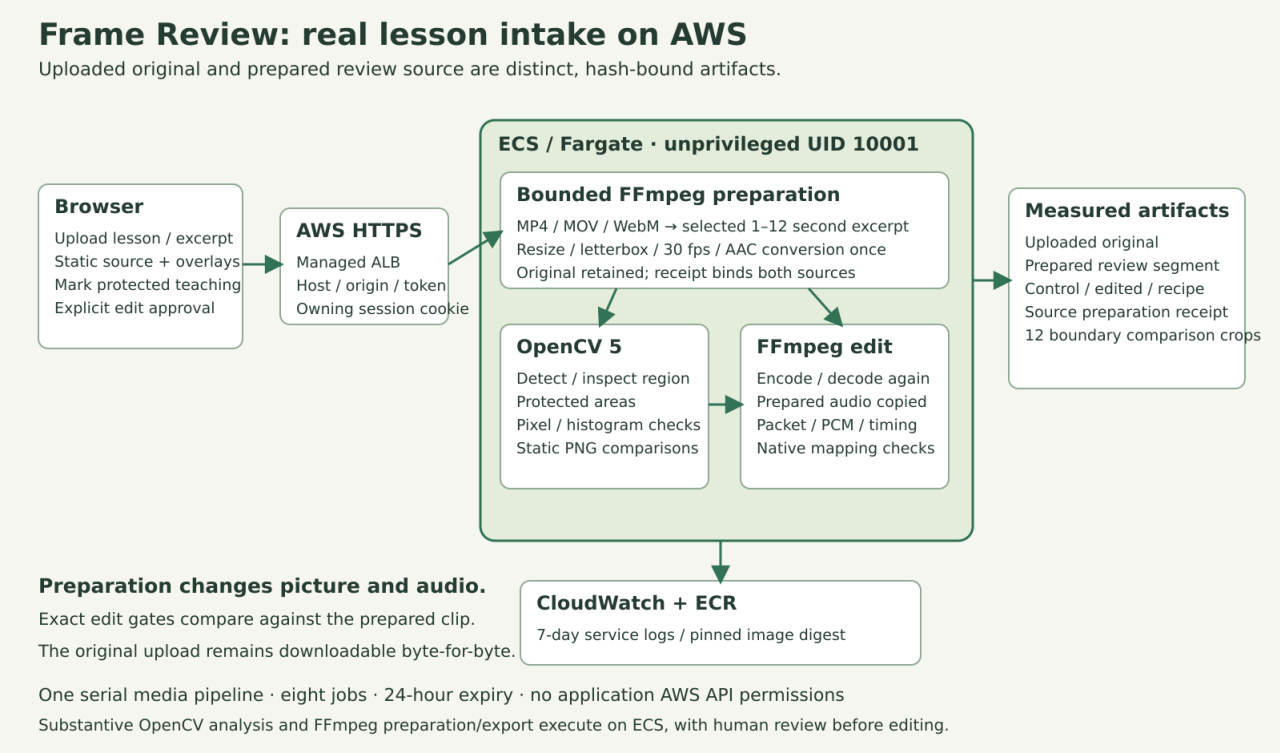
Prepared excerpt from TEFL & TESOL Courses - ITTT, Present Simple - Teaching Ideas (2013), CC BY 3.0, via Wikimedia Commons. Excerpt/resize/format conversion disclosed. No endorsement.

The problem. A creator has a flattened lesson and no editable timeline. A repeating decoration competes with the teaching. Slowing the whole clip also retimes the teacher and narration. Frame Review proposes a local retime, asks what must remain unchanged, and checks the actual encoded result.

What works. Real MP4/MOV and WebM uploads can supply a selected 1-12 second review excerpt. Preparation converts picture/audio once; exact edit preservation then compares against that prepared source. The uploaded original remains separate. No-candidate clips can be left unchanged. The positive real-footage challenge has a deliberately added defect; it is not presented as a natural repair success.

Earlier AWS app: https://le-d72c965de52749b08fd72a2513bccce1.ecs.us-east-1.on.aws
Real-lesson release source: https://github.com/josepha-mayo/opencv-frame-review

A preservation contract before an edit



Original and prepared sources. An isolated FFmpeg subprocess extracts an explicit segment, scales/letterboxes to 640x360, converts to 30 fps and encodes AAC-LC mono/48 kHz once. A receipt records original/prepared hashes, metadata, interval and transformations. Both sources remain downloadable. This is not lossless normalization.

Substantive OpenCV 5. The detector measures decoded-pixel luminance changes and candidate geometry. OpenCV also supports overlays, masked image differences, histograms and actual encoded PNG comparisons. NumPy represents native planes and mappings; FFmpeg decodes, encodes and checks media/audio. This is deterministic vision, not a trained semantic model or autonomous agent.

Consent and exact checks. Preparation never approves an edit. Consent binds source, prediction and recipe hashes. Fixed protected rectangles require zero RGB error across every frame versus the prepared source. Native outside samples/mapping, timestamps, unchanged masters and supported audio packets/timing/skip metadata/PCM are checked. Failed preservation or overlapping protection withholds the edit. Periodicity budgets can admit small variations; entry/exit transitions remain for review.

Real footage: results and negative cases

Input	Preparation / outcome
ITTT classroom: 854x480, 29.97 fps, VP8/Vorbis stereo; 218.154 s	Seconds 40-52 prepared; no candidate; explicitly left unchanged. Original-upload download exact. Narration correlation 0.99604; best measured lag zero.
Jimmyjrg screencast: 1920x1080, 60 fps, AV1/Vorbis stereo; 358.202 s	Seconds 40-52 prepared; no candidate; explicitly left unchanged. Original-upload download exact. Narration correlation 0.99975; best measured lag zero.
Real classroom plus an authored flashing corner in its black margin	Detected [588,260,32,32]. Explicit approval, 5x retime; all 360 teacher/whiteboard RGB frames exact. Native mapping/outside and prepared narration PCM exact; all twelve PNG response hashes match.

Scope. The two original excerpts were not naturally defective cases and produced no candidate. The positive case adds a controlled artifact to real teacher footage and human speech. This tests functional intake/preservation, not success repairing naturally defective lessons or creator benefit. Narration correlations use independently decoded mono/48 kHz audio and a +/-10 ms lag search; preparation remains lossy.

Frozen regression. Ten stages passed in the exact image: original units, duration exports/refusals, original audio/silent media, audio/pixel refusals, authored math lesson, real-classroom/augmented challenge, and studio UI/HTTP checks. Backend units: 137. Studio: 61 passing plus one Windows-specific Linux skip. Windows: 61 passing plus one Linux-file-check skip. Shared checks overlap; these are not independent user experiments.

Actual browser checks, local Docker adapter. The final candidate accepts the original screencast upload and returns its original bytes separately. Its real-teacher challenge completes through the browser. The edited download matches both its report and the independent export. All twelve crops load, reload recovers the completed decision, unowned known-job requests return 404, and playback reaches 12 seconds without a decoder error. This does not establish the new release on AWS.

Failures retained. A Windows bind mount hit an unchanged worker limit. A first preparation encode omitted color tags and the canonical gate refused it; explicit color parameters fixed it. UI tests caught wording drift; actual browser checks caught misleading no-edit headings and the earlier strict-CSP overlay bug. An overnight runner reset interrupted qualification and was not counted as a pass. Historical failed recipes and the missing historical screenshot remain separate.

Reproduction, AWS and remaining limits

Reproduce. Build the pinned Dockerfile; run python studio/run.py --verify /tmp/qualification on native Linux storage, then copy receipts out. The classroom original, attribution and controlled challenge are included. The external screencast is reproducible with the hash-checking fetch script and integration_real_lessons.py. The package includes source manifests, exact dependencies/binary hashes, refusal cases, provenance, architecture and deployment instructions.

AWS status. The earlier authored-lesson release remains live on ECS/Fargate behind managed HTTPS; its image is in ECR and CloudWatch retains logs seven days. The locally qualified real-lesson candidate has not replaced it: existing-account credentials expired while Joseph was away. The deployment adapter retains one serial pipeline, eight jobs, owning Secure/HttpOnly/SameSite cookies and host/origin/token checks. The new image adds UID 10001, bounded preparation and file/pipe-only demuxing, with no application AWS API role. These changes passed local tests; final cloud execution and new-release browser checks are pending. Jobs expire after 24 hours and task replacement can clear them sooner.

Admission. Upload maximum 64 MiB; one 8-bit SDR video track up to 3840x2160 and at most one supported mono/stereo audio track; source duration up to 30 minutes. Choose a start within the first 600 seconds and a 1-12 second excerpt. This release exports the segment and does not splice it back into a full lesson. HDR/high-bit-depth, subtitles/data tracks and unsupported codecs are refused. The codec-family contract exceeds the combinations tested empirically.

Impact still to establish. Test held-out naturally defective lessons with teachers/editors, against competent manual editing: task completion, review burden, unintended changes and retained teaching timing. Learning outcomes and time savings remain unmeasured. Fixed rectangles do not track objects or prove readability. Known package-scan findings remain; mitigations are not a security certification. No safety, medical or publishing-clearance claim is made.

Attribution. Classroom: ITTT, CC BY 3.0. Screencast: Jimmyjrg, CC BY 4.0; embedded Wikimedia material credited to Wikimedia Foundation under CC BY-SA 3.0. Changes and license links are in real-lesson-evaluation.md. No endorsement. The older 2:46 video uses disclosed neural narration and shows the earlier core workflow. Joseph requested his real voice for a later final demo.

Candidate source manifest: 9d94ffbd280ad5575e615cfb40f8cb2662dc94cb11609295ce0d9b1abcb48f74
Locally qualified Docker image ID: sha256:de87eb293b3cf2636e9fbb76099e7bed435ee54c5c6cf8ba8b92de7e40512a6b
Current live AWS source: d00b0c7753e4bd8debcee5ba2186c06d32530333ba1d9320376c41faaf529966
Real-footage output: 4b612c894c0ccb94787d51a085802ffe2fce32658bd53b8aeefc37b4252cacd0